Logarithms

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Logarithms

What are logarithms?

- A **logarithm** is the **inverse function** of an **exponential function**
  - Making x the subject of $a^x = b$ gives $x = \log_a(b)$
    - This is pronounced "log base *a* of *b*"
    - This is the **definition** of a logarithm
    - The number a is called the **base** of the logarithm
- If **no base** is given, it means the **base number is 10**
  - $\log(...)$ means $\log_{10}(...)$
    - In the past, this was a common base to use
    - The **log button** on a calculator is always base 10

How do I interpret logarithms?

- **log base *a* of *b***, $\log_a(b)$, is the **power** that you raise a to, to get b
  - This is the power required to take the small number, a, to the big number, b
  - For example, $\log_5 125$ is the power needed to get 5 to 125
    - The power needed is 3 (because $5^3 = 125$)
    - So $\log_5 125 = 3$
  - Similarly
    - $\log_5 25 = 2$
    - $\log_5 5 = 1$
    - $\log_5 1 = 0$ (as $5^0 = 1$)
  - It works for **negative powers** (reciprocals)
    - $\log_5\left(\frac{1}{5}\right) = -1$ (as $\frac{1}{5} = 5^{-1}$)
    - $\log_5\left(\frac{1}{25}\right) = \log_5\left(\frac{1}{5^2}\right) = -2$
  - It works for **fractional powers** (roots)
    - $\log_5(\sqrt{5}) = \frac{1}{2}$ (as $\sqrt{5} = 5^{\frac{1}{2}}$)
    - $\log_5(\sqrt[3]{5}) = \frac{1}{3}$
- Where possible, use a **calculator** to check

Stuck on something?

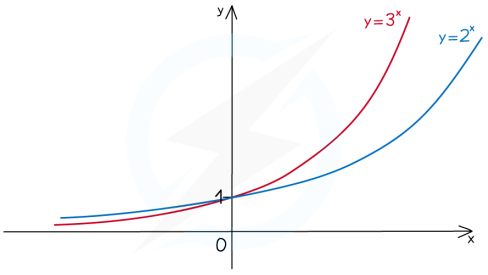
Ask questions and get help with exactly what you're revising.

Got it

10)

How do I convert between logarithms and exponentials?

- $\log_a(b) = x$ and $a^x = b$ are mathematically **identical** statements
  - You can convert between the two
- Learn to read $\log_a(b) = x$ as "x is the **power** that you raise a to, to get b"
  - This allows you to convert logs to **exponentials**
    - $a^x = b$
- This also works the other way
  - $a^x = b$ means $x = \log_a(b)$
    - Though it is often easier to think of making x the **subject**
    - To make x the subject, **take log base *a* of both sides**, $\log_a$
- This means you can relate logarithms to **exponential graphs**
  - e.g. the graph of $y = 3^x$ passes through the point $(x, 81)$
    - $3^x = 81$ so $x = \log_3 81 = 4$



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Examiner Tips and Tricks

In the exam, you can rewrite a logarithm question as an exponential question if you prefer.

How do I solve equations with logarithms?

- Equations where x is in the power are called **exponential equations**
  - Some can be solved easily **by inspection** (seeing the answer)
    - For example, $2^x = 8$ is solved by $x = 3$
  - Others cannot

- $2^x = 10$ means $x = \log_2 10$
- Type this into your **calculator** to get $x = 3.321928...$

- **Alternatively**, you can also solve equations using **logs of base 10 only**
  - The **rule** for base 10 is if $a^x = b$ then $x = \frac{\log b}{\log a}$ where $\log$ means base 10
  - For example, $2^x = 10$ means $x = \frac{\log 10}{\log 2} = 3.321928...$

Examiner Tips and Tricks

Whilst you still need to know both methods, mark schemes will accept different correct answers with logarithms, depending on whether you have used logs of base 10 or logs of a different base.

How do I use logarithms in compound interest questions?

- You can use **logarithms** to work out the power, n, in **compound interest** questions
  - e.g.you put £500 into a bank account with 5% interest per year; after how many whole years does it exceed £900?
    - $500 \times 1.05^n = 900$
    - Make n the subject: $1.05^n = \frac{900}{500} = 1.8$ so $n = \log_{1.05}(1.8)$ (or $n = \frac{\log 1.8}{\log 1.05}$)
    - This gives $n = 12.047236...$ so $n = 13$ years
    - (it has not **exceeded** 900 after 12 years)

How do I find inverse functions using logarithms?

- For **exponential functions**, you can use **logarithms** to find their **inverse functions**
  - Use the standard process for finding inverse functions
  - e.g. $f(x) = 2^{4x}$
    - $y = 2^{4x}$ then swap x and y to get $x = 2^{4y}$
    - Make $4y$ the subject using logarithms: $4y = \log_2 x$
    - Finally make y the subject to give $f^{-1}(x) = \frac{1}{4}\log_2 x$

significant figures.

Method 1

To make x the subject, take log base 4 of both sides

$$x = \log_4(20)$$

Use the $\log_{...}(...)$ button on your calculator to find x

$$x = 2.1609640...$$

Round to 3 significant figures

$x = 2.16$ **to 3 s.f.**

Method 2

Using logs of base 10, solve $a^x = b$ by using $x = \dfrac{\log b}{\log a}$

$$x = \frac{\log 20}{\log 4}$$

Use the log button on your calculator to find x

$$x = 2.1609640...$$

Round to 3 significant figures

$x = 2.16$ **to 3 s.f.**

(b) Without using a calculator, solve the equation $\log(8x) = 4$

log with no base means log base 10

$$\log_{10}(8x) = 4$$

Rewrite this statement as a exponential relationship
4 is the power that you raise 10 to, to get $8x$

$$10^4 = 8x$$

Simplify and make x the subject

$$10\,000 = 8x$$
$$\frac{10\,000}{8} = x$$

$x = 1250$

(c) Make t the subject of $7^{p(t-1)} = 2q$

Take log base 7 of both sides

$$p(t-1) = \log_7(2q)$$

Rearrange to make t the subject
Start by dividing both sides by p

$$t - 1 = \frac{1}{p}\log_7(2q)$$

Then add 1 to both sides

$$t = 1 + \frac{1}{p}\log_7(2q)$$

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